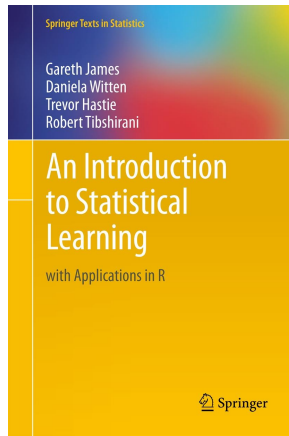


Section 1B. Course Admin

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Topics Covered in the course

1. Introduction
2. Statistical Learning
3. Linear Regression
4. Classification
5. Resampling Methods
6. Model Selection and Regularization
7. Tree-Based Methods
8. Support Vector Machines
9. Unsupervised Learning



ISL book (free pdf available)

Types of Learning Problems

Supervised learning:

- ▶ **Given** a training dataset $D = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, where
 - ▶ $\mathbf{x}_i$ are p -dimensional inputs (aka features, regressors, covariates)
 - ▶ y_i are outcome variables (aka response, target, dependent variable)
- ▶ **Find** a function able to predict outputs from inputs
 - ▶ when y_i are quantitative, we have a *regression problem*
 - ▶ when y_i are qualitative we have a *classification problem*
- ▶ **Examples:** Digit recognition, speech recognition, image classification, etc.

Unsupervised learning:

- ▶ **Given** a collection of inputs $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, without outputs
- ▶ **Find** groups of variables that behave similarly
- ▶ **Example:** Clustering problems in which we are not given labels for inputs in our dataset

Prerequisites and Workload

Necessary skills:

- ▶ Basic programming skills, preferably in Python
- ▶ A solid undergraduate probability course

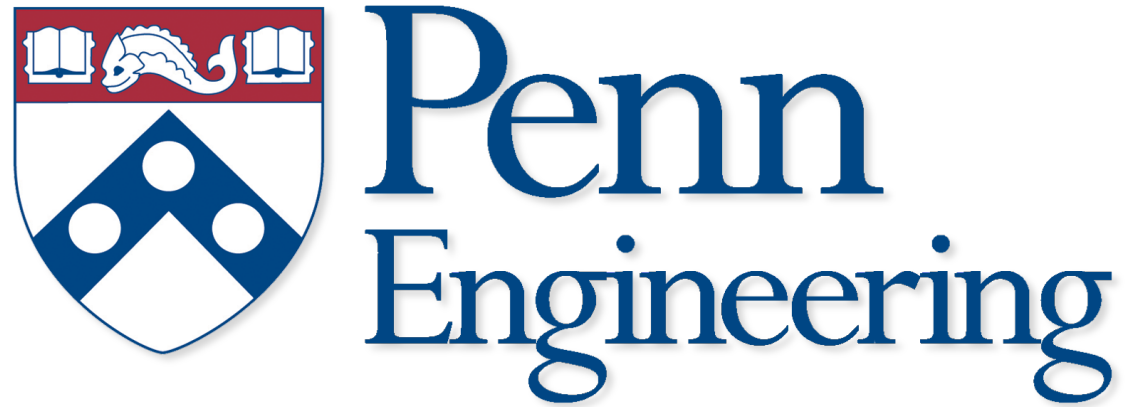
Workload:

- ▶ Watch a series of videos with theoretical content and read related sections from ISL
- ▶ Solve (graded) *assessment questions*
- ▶ At the end of each chapter, watch a video connecting the course content to Python
- ▶ Solve (graded) coding assessment in the form of Python projects

Note: Although we provide you with the Python basics to solve your projects, this is not a programming course. We rely on your previous coding experience (even if it is with a different programming language)

Grading

- ▶ Quizzes 30% (one quiz for each theoretical video)
- ▶ Programming projects 20% (one project for each coding video, at the end of each chapter)
- ▶ One midterm 25% (see Canvas site for details, e.g., date, conditions, etc.)
- ▶ One final 25% (see Canvas site for details, e.g., date, conditions, etc.)



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